



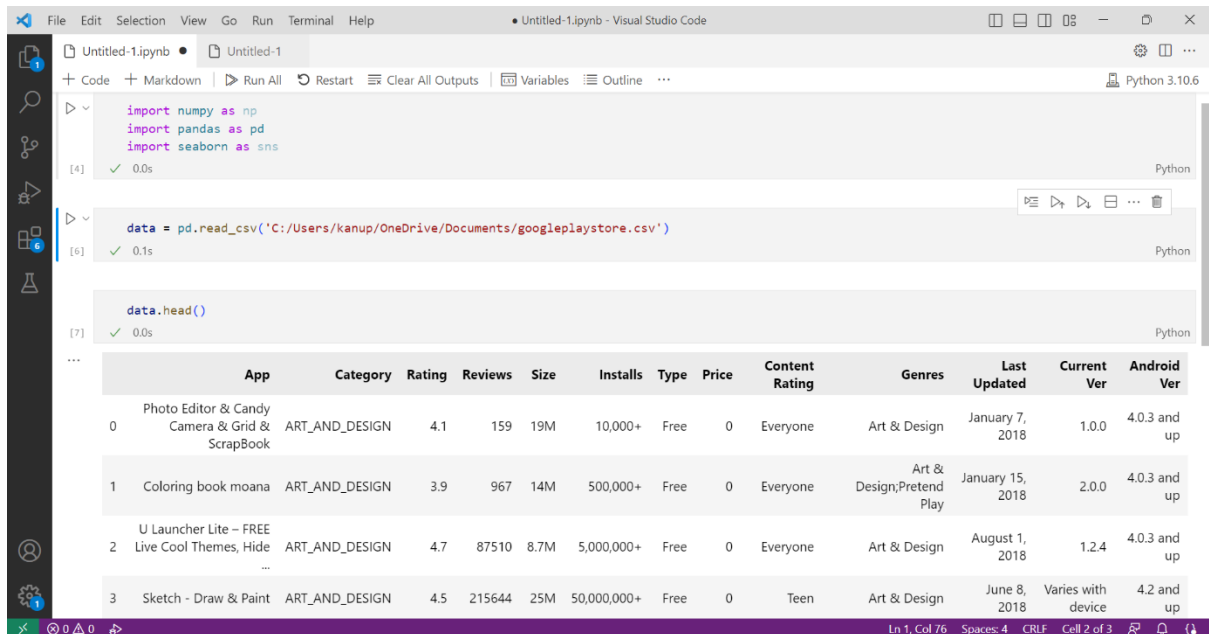
IDENTIFYING AND RECOMMENDING BEST RESTAURANT



DATA ANALYST PYTHON PROJECT

BY KANUPRIYA SINGH

Attaching screenshots of my code and its output.



The screenshot shows a Visual Studio Code window with a Jupyter Notebook titled 'Untitled-1.ipynb'. The code in the notebook is as follows:

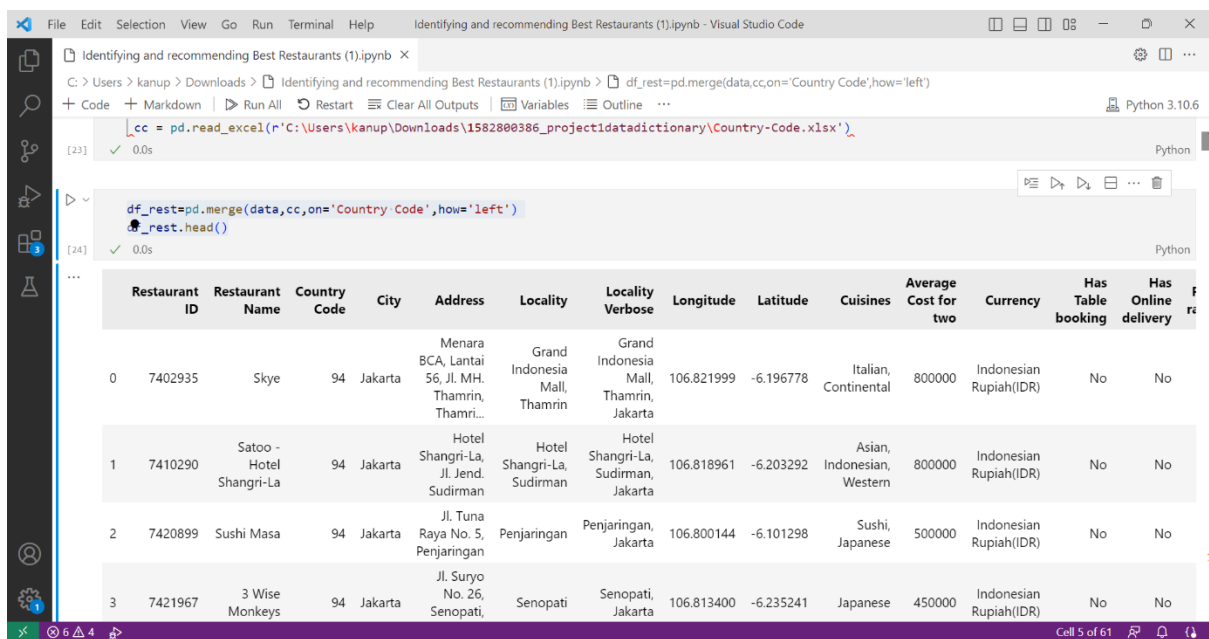
```
import numpy as np
import pandas as pd
import seaborn as sns

data = pd.read_csv('C:/Users/kanup/OneDrive/Documents/googleplaystore.csv')

data.head()
```

The output of the `data.head()` command is a table with 14 columns: App, Category, Rating, Reviews, Size, Installs, Type, Price, Content Rating, Genres, Last Updated, Current Ver, and Android Ver. The first four rows of data are shown:

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	Android Ver
0	Photo Editor & Candy Camera & Grid & ScrapBook	ART_AND_DESIGN	4.1	159	19M	10,000+	Free	0	Everyone	Art & Design	January 7, 2018	1.0.0	4.0.3 and up
1	Coloring book moana	ART_AND_DESIGN	3.9	967	14M	500,000+	Free	0	Everyone	Art & Design;Pretend Play	January 15, 2018	2.0.0	4.0.3 and up
2	U Launcher Lite – FREE Live Cool Themes, Hide ...	ART_AND_DESIGN	4.7	87510	8.7M	5,000,000+	Free	0	Everyone	Art & Design	August 1, 2018	1.2.4	4.0.3 and up
3	Sketch - Draw & Paint	ART_AND_DESIGN	4.5	215644	25M	50,000,000+	Free	0	Teen	Art & Design	June 8, 2018	Varies with device	4.2 and up



The screenshot shows a Visual Studio Code window with a Jupyter Notebook titled 'Identifying and recommending Best Restaurants (1).ipynb'. The code in the notebook is as follows:

```
cc = pd.read_excel(r'C:\Users\kanup\Downloads\1582800386_project1\data\dictionary\Country-Code.xlsx')

df_rest=pd.merge(data,cc,on='Country Code',how='left')
df_rest.head()
```

The output of the `df_rest.head()` command is a table with 15 columns: Restaurant ID, Restaurant Name, Country Code, City, Address, Locality, Locality Verbose, Longitude, Latitude, Cuisines, Average Cost for two, Currency, Has Table booking, and Has Online delivery. The first four rows of data are shown:

Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	Cuisines	Average Cost for two	Currency	Has Table booking	Has Online delivery	
0	7402935	Skye	94	Jakarta	Menara BCA, Lantai 56, Jl. MH. Thamrin, Thamrin...	Grand Indonesia Mall, Thamrin	Grand Indonesia Mall, Thamrin, Jakarta	106.821999	-6.196778	Italian, Continental	800000	Indonesian Rupiah(IDR)	No	No
1	7410290	Satoo - Hotel Shangri-La	94	Jakarta	Hotel Shangri-La, Jl. Jend. Sudirman	Hotel Shangri-La, Sudirman	Hotel Shangri-La, Sudirman, Jakarta	106.818961	-6.203292	Asian, Indonesian, Western	800000	Indonesian Rupiah(IDR)	No	No
2	7420899	Sushi Masa	94	Jakarta	Jl. Tuna Raya No. 5, Penjaringan	Penjaringan	Penjaringan, Jakarta	106.800144	-6.101298	Sushi, Japanese	500000	Indonesian Rupiah(IDR)	No	No
3	7421967	3 Wise Monkeys	94	Jakarta	Jl. Suryo No. 26, Senopati,	Senopati	Senopati, Jakarta	106.813400	-6.235241	Japanese	450000	Indonesian Rupiah(IDR)	No	No

```
File Edit Selection View Go Run Terminal Help Identifying and recommending Best Restaurants (1).ipynb - Visual Studio Code
C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > df_rest.columns = df_rest.columns.str.replace(' ','_')
+ Code + Markdown + Run All + Restart + Clear All Outputs + Variables + Outline ... Python 3.10.6

df_rest.columns = df_rest.columns.str.replace(' ','_')
df_rest.columns

[25] ✓ 0.0s Python

Index(['Restaurant_ID', 'Restaurant_Name', 'Country_Code', 'City', 'Address',
      'Locality', 'Locality_Verbose', 'Longitude', 'Latitude', 'Cuisines',
      'Average_Cost_for_two', 'Currency', 'Has_Table_booking',
      'Has_Online_delivery', 'Price_range', 'Aggregate_rating',
      'Rating_color', 'Rating_text', 'Votes', 'Country'],
      dtype='object')

df_rest.info()

[26] ✓ 0.0s Python

<class 'pandas.core.frame.DataFrame'>
Int64Index: 9551 entries, 0 to 9550
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   Restaurant_ID          9551 non-null  int64
1   Restaurant_Name        9550 non-null  object
2   Country_Code           9551 non-null  int64
3   City                   9551 non-null  object
4   Address                9551 non-null  object
5   Locality               9551 non-null  object
6   Locality_Verbose       9551 non-null  object
```

```
File Edit Selection View Go Run Terminal Help Identifying and recommending Best Restaurants (1).ipynb - Visual Studio Code
C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > df_rest.columns = df_rest.columns.str.replace(' ','_')
+ Code + Markdown + Run All + Restart + Clear All Outputs + Variables + Outline ... Python 3.10.6

df_rest.isnull().sum() #total number of null entries per column

[27] ✓ 0.0s Python

Restaurant_ID      0
Restaurant_Name    1
Country_Code       0
City               0
Address            0
Locality           0
Locality_Verbose   0
Longitude          0
Latitude           0
Cuisines           9
Average_Cost_for_two 0
Currency           0
Has_Table_booking  0
Has_Online_delivery 0
Price_range        0
Aggregate_rating    0
Rating_color        0
Rating_text         0
Votes              0
Country            0
dtype: int64
```

Identifying and recommending Best Restaurants (1).ipynb - Visual Studio Code

File Edit Selection View Go Run Terminal Help

Identifying and recommending Best Restaurants (1).ipynb

C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > df_rest[df_rest['Restaurant_Name'].isnull()]

+ Code + Markdown | Run All | Restart | Clear All Outputs | Variables | Outline

Python 3.10.6

```
df_rest[df_rest['Restaurant_Name'].isnull()]
```

[28] ✓ 0.0s Python

Restaurant_ID	Restaurant_Name	Country_Code	City	Address	Locality	Locality_Verbose	Longitude	Latitude	Cuisines	Average_Cost_for_1
1646	113702	NaN	1 Ahmedabad	Opposite Sindhu Bhawan, Bodakdev, Ahmedabad	Bodakdev	Bodakdev, Ahmedabad	72.501764	23.040163	North Indian, Continental, Mexican, Italian	

```
#Since the restaurant name is missing, we dropped the record and reset the index.
df_rest.dropna(axis=0,subset=['Restaurant_Name'],inplace=True)
df_rest.reset_index(drop=True,inplace=True)
df_rest[df_rest['Cuisines'].isnull()]
```

[29] ✓ 0.0s Python

Restaurant_ID	Restaurant_Name	Country_Code	City	Address	Locality	Locality_Verbose	Longitude	Latitude	Cuisines	Average_Cost_for_two
9082	17374552	Corkscrew Cafe	216 Gainesville	51 W Main St, Dahlonaga, GA 30533	Dahlonaga	Dahlonaga, Gainesville	-83.985800	34.531800	NaN	40

Identifying and recommending Best Restaurants (1).ipynb - Visual Studio Code

File Edit Selection View Go Run Terminal Help

Identifying and recommending Best Restaurants (1).ipynb

C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > df_rest[df_rest['Restaurant_Name'].isnull()]

+ Code + Markdown | Run All | Restart | Clear All Outputs | Variables | Outline

Python 3.10.6

```
#Since there were only 9 records without cuisines, we have replace the null values with Others.
df_rest['Cuisines'].fillna('Others',inplace=True)
```

[30] ✓ 0.0s Python

```
df_rest.isnull().sum()
df_rest.info()
```

[31] ✓ 0.0s Python

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9550 entries, 0 to 9549
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Restaurant_ID          9550 non-null   int64
1   Restaurant_Name        9550 non-null   object
2   Country_Code           9550 non-null   int64
3   City                   9550 non-null   object
4   Address                 9550 non-null   object
5   Locality                9550 non-null   object
6   Locality_Verbose       9550 non-null   object
7   Longitude              9550 non-null   float64
8   Latitude               9550 non-null   float64
9   Cuisines                9550 non-null   object
10  Average_Cost_for_two    9550 non-null   int64
11  Currency                9550 non-null   object
12  Has_Table_booking       9550 non-null   object
```

Identifying and recommending Best Restaurants (1).ipynb - Visual Studio Code

Identifying and recommending Best Restaurants (1).ipynb ×

C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > df_rest[df_rest['Restaurant_Name'].isnull()]

+ Code + Markdown | Run All | Restart | Clear All Outputs | Variables | Outline ... Python 3.10.6

```
#Explore the geographical distribution of the restaurants.
#Finding out the cities with maximum / minimum number of restaurants
#Explore the franchise with most national presence
#Ratio between restaurants that allow table booking vs that do not allow table booking.
#Percentage of restaurants providing online delivery.
#Difference in no. of votes for the restaurants that deliver and the restaurant that don't.
```

[32] ✓ 0.0s Python

```
centry_dist = df_rest.groupby(['Country_Code','Country']).agg( Count = ('Restaurant_ID','count'))
centry_dist.sort_values(by='Count',ascending=False)
```

[33] ✓ 0.0s Python

... We observe that India has then highest number of restaurants with 8651 restaurants and USA is number 2 with 434 restaurants

Country_Code	Country	Count
1	India	8651
216	United States	434
215	United Kingdom	80
30	Brazil	60
189	South Africa	60

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Identifying and recommending Best Restaurants (1).ipynb - Visual Studio Code

Identifying and recommending Best Restaurants (1).ipynb ×

C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > df_rest[df_rest['Restaurant_Name'].isnull()]

+ Code + Markdown | Run All | Restart | Clear All Outputs | Variables | Outline ... Python 3.10.6

```
centry_dist.plot(kind='barh')
```

[34] ✓ 0.6s Python

... <AxesSubplot: ylabel='Country_Code,Country'>

</>

Country_Code,Country

Count

Country_Code,Country	Count
(216, United States)	434
(215, United Kingdom)	80
(214, UAE)	60
(208, Turkey)	60
(191, Sri Lanka)	60
(189, South Africa)	60
(184, Singapore)	60
(166, Qatar)	60
(162, Philippines)	60
(148, New Zealand)	60
(94, Indonesia)	60
(37, Canada)	60
(30, Brazil)	60
(14, Australia)	60
(1, India)	8651

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29°C

08:15 PM

```
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+ Code + Markdown Run All Restart Clear All Outputs Variables Outline Python 3.10.6

city_dist = df_rest.groupby(['Country','City']).agg(Count = ('Restaurant_ID','count'))
city_dist.describe()
#city with max restaurant has count = 5473
#city with min restaurant has count = 1

[35] ✓ 0.0s Python

Count
count 141.000000
mean 67.730496
std 476.723952
min 1.000000
25% 1.000000
50% 20.000000
75% 20.000000
max 5473.000000

city_dist.sort_values(by='Count',ascending=False)
# we see that new Delhi has the maximum restaurant with 5473
# we observe that multiple cities have only one restaurant.

[36] ✓ 0.0s Python

Count
```

```
File Edit Selection View Go Run Terminal Help Identifying and recommending Best Restaurants (1).ipynb - Visual Studio Code
C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > city_dist.sort_values(by='Count',ascending=False)
+ Code + Markdown Run All Restart Clear All Outputs Variables Outline Python 3.10.6

city_dist.sort_values(by='Count',ascending=False)
# we see that new Delhi has the maximum restaurant with 5473
# we observe that multiple cities have only one restaurant.

[36] ✓ 0.0s Python

Count
Country City
India New Delhi 5473
Gurgaon 1118
Noida 1080
Faridabad 251
Ghaziabad 25
... ..
Panchkula 1
Australia Balingup 1
Indonesia Bandung 1
Phillipines Quezon City 1
United States Winchester Bay 1

141 rows x 1 columns
```

```
File Edit Selection View Go Run Terminal Help Identifying and recommending Best Restaurants (1).ipynb - Visual Studio Code
C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > min_cnt_rest = city_dist[city_dist['Count']==1]
+ Code + Markdown | Run All | Restart | Clear All Outputs | Variables | Outline ... Python 3.10.6
```

```
[37] ✓ 0.0s
min_cnt_rest = city_dist[city_dist['Count']==1]
min_cnt_rest.info()
min_cnt_rest
#There are 46 cities in 7 different countries with 1 restaurants

<class 'pandas.core.frame.DataFrame'>
MultiIndex: 46 entries, ('Australia', 'Armidale') to ('United States', 'Winchester Bay')
Data columns (total 1 columns):
 # Column Non-Null Count Dtype
---
 0 Count 46 non-null int64
dtypes: int64(1)
memory usage: 1.8+ KB
```

Country	City	Count
Australia	Armidale	1
	Balingup	1
	Beechworth	1
	Dicky Beach	1
	East Ballina	1
	Flaxton	1

```
File Edit Selection View Go Run Terminal Help Identifying and recommending Best Restaurants (1).ipynb - Visual Studio Code
C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > franch_dist = df_rest.groupby(['Restaurant_Name','Country']).agg(Count = ('Country','count'))
+ Code + Markdown | Run All | Restart | Clear All Outputs | Variables | Outline ... Python 3.10.6
```

```
[38] ✓ 0.0s
franch_dist = df_rest.groupby(['Restaurant_Name','Country']).agg(Count = ('Country','count'))
franch_dist.describe()
```

Count
count 7472.000000
mean 1.278105
std 2.165675
min 1.000000
25% 1.000000
50% 1.000000
75% 1.000000
max 83.000000

```
franch_dist.sort_values(by='Count',ascending=False)
#cafe Coffe day has most national prsesence
```

```
[39] ✓ 0.0s
```

Restaurant Name	Country	Count
Cafe Coffee Day	India	83

Identifying and recommending Best Restaurants (1).ipynb - Visual Studio Code

Identifying and recommending Best Restaurants (1).ipynb

C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > `franch_dist = df_rest.groupby(['Restaurant_Name','Country']).agg(Count = ('Country','count'))`

+ Code + Markdown | Run All | Restart | Clear All Outputs | Variables | Outline | Python 3.10.6

```
franch_dist.sort_values(by='Count',ascending=False)
#cafe Coffe day has most national prsesence
```

[39] ✓ 0.0s Python

...

Restaurant_Name	Country	Count
Cafe Coffee Day	India	83
Domino's Pizza	India	79
Subway	India	63
Green Chick Chop	India	51
McDonald's	India	48
...	...	...
Gibson's Gourmet Burgers & Ribs	South Africa	1
Giapo	New Zealand	1
Giani's di Hatti	India	1
Gian Ji Punjabi Dhaba	India	1
İaukura€Ua Sofras€±	Turkey	1

7472 rows x 1 columns

Identifying and recommending Best Restaurants (1).ipynb - Visual Studio Code

Identifying and recommending Best Restaurants (1).ipynb

C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > `#Ratio between restaurants allowing table booking and those which dont`

+ Code + Markdown | Run All | Restart | Clear All Outputs | Variables | Outline | Python 3.10.6

```
dummy = ['Has_Table_booking','Has_Online_delivery']
df_rest1 = pd.get_dummies(df_rest1,columns=dummy,drop_first=True)
df_rest1.head()
# 0 indicates 'NO'
# 1 indicates 'YES'
```

[41] ✓ 0.0s Python

...

sines	Average_Cost_for_two	Currency	Price_range	Aggregate_rating	Rating_color	Rating_text	Votes	Country	Has_Table_booking_Yes	Has_Online_delivery_Yes
alian, ental	800000	Indonesian Rupiah(IDR)	3	4.1	Green	Very Good	1498	Indonesia	0	0
isian, isian, stern	800000	Indonesian Rupiah(IDR)	3	4.6	Dark Green	Excellent	873	Indonesia	0	0
iushi, inese	500000	Indonesian Rupiah(IDR)	3	4.9	Dark Green	Excellent	605	Indonesia	0	0
inese	450000	Indonesian Rupiah(IDR)	3	4.2	Green	Very Good	395	Indonesia	0	0

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```
File Edit Selection View Go Run Terminal Help • Identifying and recommending Best Restaurants (1).ipynb - Visual Studio Code

C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > #Ratio between restaurants allowing table booking and those which dont
+ Code + Markdown | Run All | Restart | Clear All Outputs | Variables | Outline ... Python 3.10.6

#Ratio between restaurants allowing table booking and those which dont
table_booking = df_rest1[df_rest1['Has_Table_booking_Yes']==1]['Restaurant_ID'].count()
table_nbooking = df_rest1[df_rest1['Has_Table_booking_Yes']==0]['Restaurant_ID'].count()
print('Ratio between restaurants that allow table booking vs. those that do not allow table booking: ',
      round((table_booking/table_nbooking),2))

[42] ✓ 0.0s Python

... Ratio between restaurants that allow table booking vs. those that do not allow table booking: 0.14

print(table_booking,table_nbooking)

[43] ✓ 0.0s Python

... 1158 8392

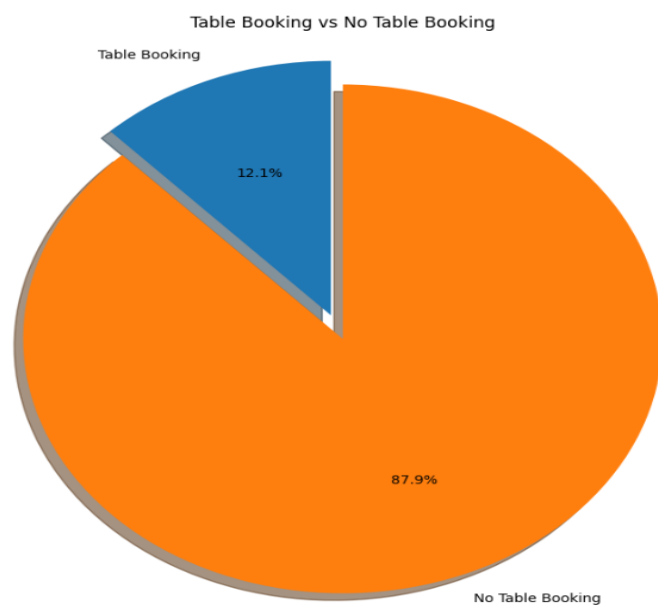
#Pie chart to show percentage of restaurants which allow table booking and those which don't
labels = 'Table Booking', 'No Table Booking'
sizes = [table_booking,table_nbooking]
explode = (0.1, 0) # only "explode" the 2nd slice (i.e. 'Hogs')

fig1, ax1 = plt.subplots(figsize=(9,9))
ax1.pie(sizes, explode=explode, labels=labels, autopct='%1.1f%%',shadow=True, startangle=90)
ax1.set_title("Table Booking vs No Table Booking")
ax1.axis('equal') # Equal aspect ratio ensures that pie is drawn as a circle.

fig1, ax1 = plt.subplots(figsize=(9,9))
ax1.pie(sizes, explode=explode, labels=labels, autopct='%1.1f%%',shadow=True, startangle=90)
ax1.set_title("Table Booking vs No Table Booking")
ax1.axis('equal') # Equal aspect ratio ensures that pie is drawn as a circle.

plt.show()

[44] ✓ 0.2s Python
```



File Edit Selection View Go Run Terminal Help • Identifying and recommending Best Restaurants (1).ipynb - Visual Studio Code

Identifying and recommending Best Restaurants (1).ipynb • Plots

C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > #Percentage of restaurant that has online delivery

+ Code + Markdown | ▶ Run All ⌂ Restart ☰ Clear All Outputs | 📄 Variables 📄 Outline ... Python 3.10.6

```
#Percentage of restaurant that has online delivery
rest_od = df_rest1[df_rest1['Has_Online_delivery_Yes'] == 1]['Restaurant_ID'].count()
rest_nod = df_rest1[df_rest1['Has_Online_delivery_Yes'] == 0]['Restaurant_ID'].count()
print('Percentage of restaurants providing online delivery : {} %'.format((round(rest_od/len(df_rest1),3)*100)))
```

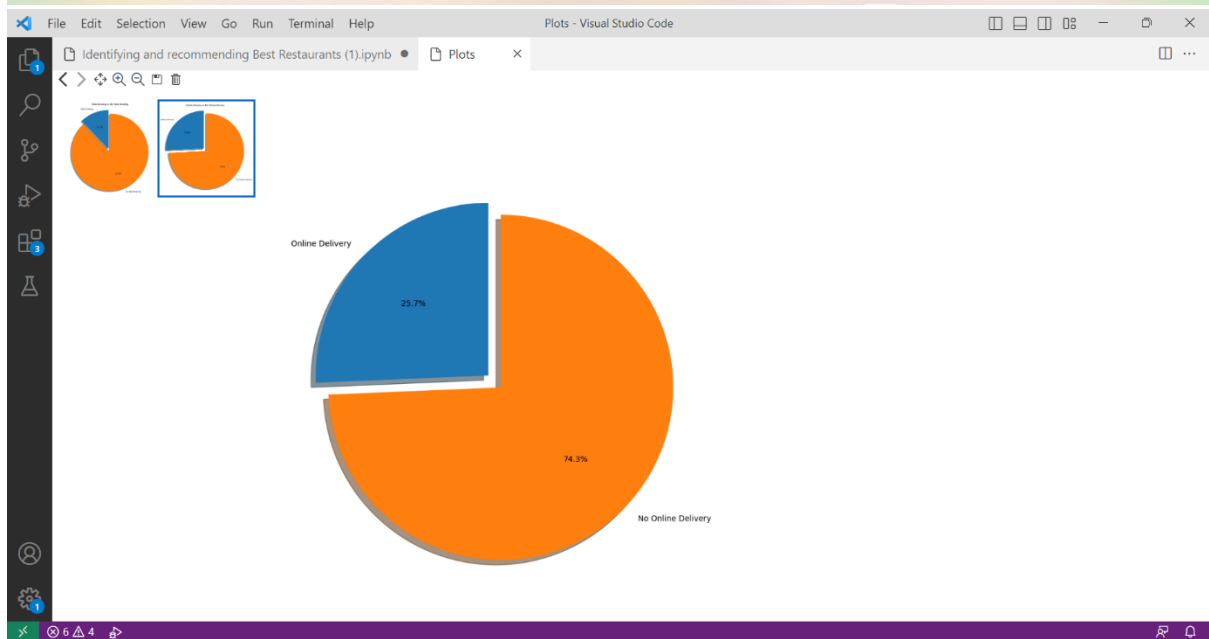
[45] ✓ 0.0s Python

Percentage of restaurants providing online delivery : 25.7 %

```
#pie chart to show percentages of restaurants allowing online delivery vs those which do not have online delivery
labels = 'Online Delivery','No Online Delivery'
size = [rest_od,rest_nod]
explode = (0.1,0)
fig1,ax1 = plt.subplots(figsize=(9,9))
ax1.pie(size,explode=explode,labels=labels,autopct='%1.1f%%',shadow=True,startangle=90)
ax1.set_title("Online Delivery vs No Online Delivery")
ax1.axis('equal')
plt.show()
```

[46] ✓ 0.1s Python

Online Delivery vs No Online Delivery



File Edit Selection View Go Run Terminal Help

Identifying and recommending Best Restaurants (1).ipynb - Visual Studio Code

Identifying and recommending Best Restaurants (1).ipynb • Plots

C:\Users> kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > #pie chart to show percentages of restaurants allowing online delivery vs those which do not have online del

+ Code + Markdown | Run All Restart Clear All Outputs Variables Outline ... Python 3.10.6

```
rest_deliver = df_rest1[df_rest1['Has_Table_booking_Yes'] == 1]['Votes'].sum()
rest_ndeliver = df_rest1[df_rest1['Has_Table_booking_Yes'] == 0]['Votes'].sum()
print('Difference in number of votes for restaurants that deliver and dont deliver: ',abs((rest_deliver - rest_ndeliver)))
```

[47] ✓ 0.0s Python

... Difference in number of votes for restaurants that deliver and dont deliver: 680082

```
#What are the top 10 cuisines served across cities?
#Explore how ratings are distributed overall
#Evaluate the Highest Rated and Lowest Rated Restaurant of the City in all the countries
#What is the maximum and minimum no. of cuisines that a restaurant serves?
```

[48] ✓ 0.0s Python

```
df_rest.columns
cuisines = df_rest['Cuisines'].apply(lambda x: pd.Series(x.split(',')))
```

[49] ✓ 3.0s Python

```
cuisines.columns = ['Cuisine_1','Cuisine_2','Cuisine_3','Cuisine_4','Cuisine_5','Cuisine_6','Cuisine_7','Cuisine_8']
cuisines.tail()
```

Ln 10, Col 1 Spaces: 4 CRLF Cell 27 of 61 Python 3.10.6

File Edit Selection View Go Run Terminal Help

Identifying and recommending Best Restaurants (1).ipynb - Visual Studio Code

Identifying and recommending Best Restaurants (1).ipynb • Plots

C:\Users> kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > #pie chart to show percentages of restaurants allowing online delivery vs those which do not have online del

+ Code + Markdown | Run All Restart Clear All Outputs Variables Outline ... Python 3.10.6

```
cuisines.columns = ['Cuisine_1','Cuisine_2','Cuisine_3','Cuisine_4','Cuisine_5','Cuisine_6','Cuisine_7','Cuisine_8']
cuisines.tail()
```

[50] ✓ 0.0s Python

...

	Cuisine_1	Cuisine_2	Cuisine_3	Cuisine_4	Cuisine_5	Cuisine_6	Cuisine_7	Cuisine_8
9545	Chinese	North Indian	Fast Food	NaN	NaN	NaN	NaN	NaN
9546	Indian	Chinese	Continental	NaN	NaN	NaN	NaN	NaN
9547	Cafe	Continental	Desserts	Ice Cream	Italian	Beverages	NaN	NaN
9548	Street Food	NaN	NaN	NaN	NaN	NaN	NaN	NaN
9549	Chinese	North Indian	NaN	NaN	NaN	NaN	NaN	NaN

[51] ✓ 0.0s Python

...

	Restaurant_ID	Restaurant_Name	Country_Code	City	Address	Locality	Locality_Verbose	Longitude	Latitude	Cuisines	...	Votes	Country	Cu
0	7402935		Skye	94 Jakarta	Menara BCA, Lantai 56, Jl. MH. Thamrin,	Grand Indonesia Mall, Thagrin	Grand Indonesia Mall, Thamrin, Jakarta	106.821999	-6.196778	Italian, Continental	...	1498	Indonesia	

Ln 10, Col 1 Spaces: 4 CRLF Cell 27 of 61 Python 3.10.6

Identifying and recommending Best Restaurants (1).ipynb - Visual Studio Code

File Edit Selection View Go Run Terminal Help • Identifying and recommending Best Restaurants (1).ipynb - Visual Studio Code

C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > #What are the top 10 cuisines served across cities?

+ Code + Markdown | Run All | Restart | Clear All Outputs | Variables | Outline ... Python 3.10.6

```
cuisine_loc = pd.DataFrame(df_cuisines[['Country','City','Locality_Verbose','Cuisine_1','Cuisine_2','Cuisine_3',
                                         'Cuisine_4','Cuisine_5','Cuisine_6','Cuisine_7','Cuisine_8']])
```

[52] ✓ 0.0s Python

```
cuisine_loc_stack=pd.DataFrame(cuisine_loc.stack()) #stacking the columns
cuisine_loc.head()
```

[53] ✓ 0.0s Python

	Country	City	Locality_Verbose	Cuisine_1	Cuisine_2	Cuisine_3	Cuisine_4	Cuisine_5	Cuisine_6	Cuisine_7	Cuisine_8
0	Indonesia	Jakarta	Grand Indonesia Mall, Thamrin, Jakarta	Italian	Continental	NaN	NaN	NaN	NaN	NaN	NaN
1	Indonesia	Jakarta	Hotel Shangri-La, Sudirman, Jakarta	Asian	Indonesian	Western	NaN	NaN	NaN	NaN	NaN
2	Indonesia	Jakarta	Penjaringan, Jakarta	Sushi	Japanese	NaN	NaN	NaN	NaN	NaN	NaN
3	Indonesia	Jakarta	Senopati, Jakarta	Japanese	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4	Indonesia	Jakarta	Thamrin, Jakarta	French	Western	NaN	NaN	NaN	NaN	NaN	NaN

```
keys = [c for c in cuisine_loc if c.startswith('Cuisine')]
```

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Identifying and recommending Best Restaurants (1).ipynb - Visual Studio Code

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C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > rest_deliver = df_rest1[df_rest1['Has_Table_booking_Yes'] == 1]['Votes'].sum()

+ Code + Markdown | Run All | Restart | Clear All Outputs | Variables | Outline ... Python 3.10.6

```
keys = [c for c in cuisine_loc if c.startswith('Cuisine')]
a=pd.melt(cuisine_loc, id_vars='Locality_Verbose', value_vars=keys, value_name='Cuisines')
#melting the stack into one row
max_rate=pd.DataFrame(a.groupby(by=['Locality_Verbose','variable','Cuisines']).size().reset_index())
#find the highest restaurant in the city
max_rate
del max_rate['variable']
max_rate.columns=['Locality_Verbose','Cuisines','Count']
max_rate.head()
```

[54] ✓ 0.0s Python

	Locality_Verbose	Cuisines	Count
0	ILD Trade Centre Mall, Sohna Road, Gurgaon	Cafe	1
1	ILD Trade Centre Mall, Sohna Road, Gurgaon	North Indian	1
2	ILD Trade Centre Mall, Sohna Road, Gurgaon	Beverages	1
3	ILD Trade Centre Mall, Sohna Road, Gurgaon	Mughlai	1
4	12th Square Building, Banjara Hills, Hyderabad	Mughlai	1

```
#find the highest restaurant in the city
loc=max_rate.sort_values('Count', ascending=False).groupby(by=['Locality_Verbose'],as_index=False).first()
loc.head()
```

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Identifying and recommending Best Restaurants (1).ipynb - Visual Studio Code

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Identifying and recommending Best Restaurants (1).ipynb • Plots

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+ Code + Markdown | Run All | Restart | Clear All Outputs | Variables | Outline ... Python 3.10.6

```
#find the highest restaurant in the city
loc=max_rate.sort_values('Count', ascending=False).groupby(by=['Locality_Verbose'],as_index=False).first()
loc.head()
```

[55] ✓ 0.0s Python

	Locality_Verbose	Cuisines	Count
0	ILD Trade Centre Mall, Sohna Road, Gurgaon	Cafe	1
1	12th Square Building, Banjara Hills, Hyderabad	Mughlai	1
2	A Hotel, Gurdev Nagar, Ludhiana	Chinese	1
3	ARSS Mall, Paschim Vihar, New Delhi	North Indian	1
4	Aaya Nagar, New Delhi	Cuisine Varies	1

```
rating_res=loc.merge(df_rest,left_on='Locality_Verbose',right_on='Locality_Verbose',how='inner')
#inner join to merge the two dataframe
df=pd.DataFrame(rating_res[['Country','City','Locality_Verbose','Cuisines_x','Count']])
#making a dataframe of rating restaurant
country=rating_res.sort_values('Count', ascending=False).groupby(by=['Country'],as_index=False).first()
#grouping the data by country code
con=pd.DataFrame(country[['Country','City','Locality','Cuisines_x','Count']])
con.columns=['Country','City','Locality','Cuisines','Number of restaurants in the country']
#renaming the columns
con1=con.sort_values('Number of restaurants in the country', ascending=False)
#sorting the restaurants on the basis of the number of restaurants in the country
con1[:10]
final_con=con1.drop(con1.index[[7,10]])
```

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Identifying and recommending Best Restaurants (1).ipynb - Visual Studio Code

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+ Code + Markdown | Run All | Restart | Clear All Outputs | Variables | Outline ... Python 3.10.6

```
rating_res=loc.merge(df_rest,left_on='Locality_Verbose',right_on='Locality_Verbose',how='inner')
#inner join to merge the two dataframe
df=pd.DataFrame(rating_res[['Country','City','Locality_Verbose','Cuisines_x','Count']])
#making a dataframe of rating restaurant
country=rating_res.sort_values('Count', ascending=False).groupby(by=['Country'],as_index=False).first()
#grouping the data by country code
con=pd.DataFrame(country[['Country','City','Locality','Cuisines_x','Count']])
con.columns=['Country','City','Locality','Cuisines','Number of restaurants in the country']
#renaming the columns
con1=con.sort_values('Number of restaurants in the country', ascending=False)
#sorting the restaurants on the basis of the number of restaurants in the country
con1[:10]
final_con=con1.drop(con1.index[[7,10]])
```

[56] ✓ 0.0s Python

[57] ✓ 0.0s Python

	Country	City	Locality	Cuisines	Number of restaurants in the country
3	India	New Delhi	Connaught Place	North Indian	48
14	United States	Dubuque	Dubuque	American	9
5	New Zealand	Wellington City	Te Aro	Cafe	5

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Identifying and recommending Best Restaurants (1).ipynb

Plots

C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > rest_deliver = df_rest1[df_rest1['Has_Table_booking_Yes'] == 1]['Votes'].sum()

+ Code + Markdown Run All Restart Clear All Outputs Variables Outline Python 3.10.6

```

final_con

[57] ✓ 0.0s Python
...

```

	Country	City	Locality	Cuisines	Number of restaurants in the country
3	India	New Delhi	Connaught Place	North Indian	48
14	United States	Dubuque	Dubuque	American	9
5	New Zealand	Wellington City	Te Aro	Cafe	5
1	Brazil	Rio de Janeiro	Ipanema	Brazilian	3
6	Phillipines	Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City	Japanese	2
8	Singapore	Singapore	Marina Centre, Downtown Core	Seafood	2
9	South Africa	Cape Town	Green Point	Grill	2
11	Turkey	Ankara	Gazi Osman PaÅua	World Cuisine	2
12	UAE	Abu Dhabi	Abu Dhabi Mall, Tourist Club Area (Al Zahiyah)	American	2
0	Australia	Victor Harbor	Victor Harbor	Coffee and Tea	1
2	Canada	Vineland Station	Vineland Station	Italian	1
4	Indonesia	Jakarta	Tebet	Western	1
7	Qatar	Doha	The Westin Doha Hotel & Spa, Fereej Bin Mahmoud	Thai	1

```

rest_cuisine = pd.DataFrame(df_cuisines[['Restaurant_Name','City','Cuisine_1','Cuisine_2','Cuisine_3','Cuisine_4',
                                         'Cuisine_5','Cuisine_6','Cuisine_7','Cuisine_8']])

```

Ln 13, Col 1 Spaces: 4 CRLF Cell 28 of 61 Python 3.10.6

Identifying and recommending Best Restaurants (1).ipynb - Visual Studio Code

Identifying and recommending Best Restaurants (1).ipynb

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C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > rest_deliver = df_rest1[df_rest1['Has_Table_booking_Yes'] == 1]['Votes'].sum()

+ Code + Markdown Run All Restart Clear All Outputs Variables Outline Python 3.10.6

```

rest_cuisine = pd.DataFrame(df_cuisines[['Restaurant_Name','City','Cuisine_1','Cuisine_2','Cuisine_3','Cuisine_4',
                                         'Cuisine_5','Cuisine_6','Cuisine_7','Cuisine_8']])
rest_cuisine_stack=pd.DataFrame(rest_cuisine.stack()) #stacking the columns
rest_cuisine.head()

[58] ✓ 0.0s Python
...

```

	Restaurant_Name	City	Cuisine_1	Cuisine_2	Cuisine_3	Cuisine_4	Cuisine_5	Cuisine_6	Cuisine_7	Cuisine_8
0	Skye	Jakarta	Italian	Continental	NaN	NaN	NaN	NaN	NaN	NaN
1	Satoo - Hotel Shangri-La	Jakarta	Asian	Indonesian	Western	NaN	NaN	NaN	NaN	NaN
2	Sushi Masa	Jakarta	Sushi	Japanese	NaN	NaN	NaN	NaN	NaN	NaN
3	3 Wise Monkeys	Jakarta	Japanese	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4	Avec Moi Restaurant and Bar	Jakarta	French	Western	NaN	NaN	NaN	NaN	NaN	NaN

```

keys1 = [c for c in rest_cuisine if c.startswith('Cuisine')]
b=pd.melt(rest_cuisine, id_vars='Restaurant_Name', value_vars=keys, value_name='Cuisines')
#melting the stack into one row
max_rate1=pd.DataFrame(b.groupby(by=['Restaurant_Name','variable','Cuisines']).size().reset_index())
#find the highest restuarant in the city
max_rate1
del max_rate1['variable']
max_rate1.columns=['Restaurant_Name','Cuisines','Count']
max_rate1.head(20)

```

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```
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Identifying and recommending Best Restaurants (1).ipynb • Plots
C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > rest_deliver = df_rest1[df_rest1['Has_Table_booking_Yes'] == 1]['Votes'].sum()
+ Code + Markdown | Run All | Restart | Clear All Outputs | Variables | Outline ... Python 3.10.6

max_rate1.sort_values('Count',ascending=False)
#Cafe Coffee Day has the max number of cuisines and The least number of cuisines in a restaurant is 1.

[50] ✓ 0.0s Python
...

```

	Restaurant_Name	Cuisines	Count
2479	Cafe Coffee Day	Cafe	83
4596	Domino's Pizza	Pizza	79
4597	Domino's Pizza	Fast Food	78
12984	Subway	Salad	63
12985	Subway	Healthy Food	63
...	...	...	...
5568	Gabbar's Bar & Kitchen	Chinese	1
5569	Gabbar's Bar & Kitchen	Mexican	1
5570	Gabbar's Bar & Kitchen	Italian	1
5571	Gaga Manjero	World Cuisine	1
15963	İaukura€Ua Sofras€±	Izgara	1

15964 rows x 3 columns

```
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Identifying and recommending Best Restaurants (1).ipynb • Plots
C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > rest_deliver = df_rest1[df_rest1['Has_Table_booking_Yes'] == 1]['Votes'].sum()
+ Code + Markdown | Run All | Restart | Clear All Outputs | Variables | Outline ... Python 3.10.6

rating = df_rest1[['Restaurant_ID','Restaurant_Name','Country','City','Aggregate_rating','Average_Cost_for_two','Votes','Price_range','Has_Table_booking_Yes','Has_Online_delivery']]

[61] ✓ 0.0s Python

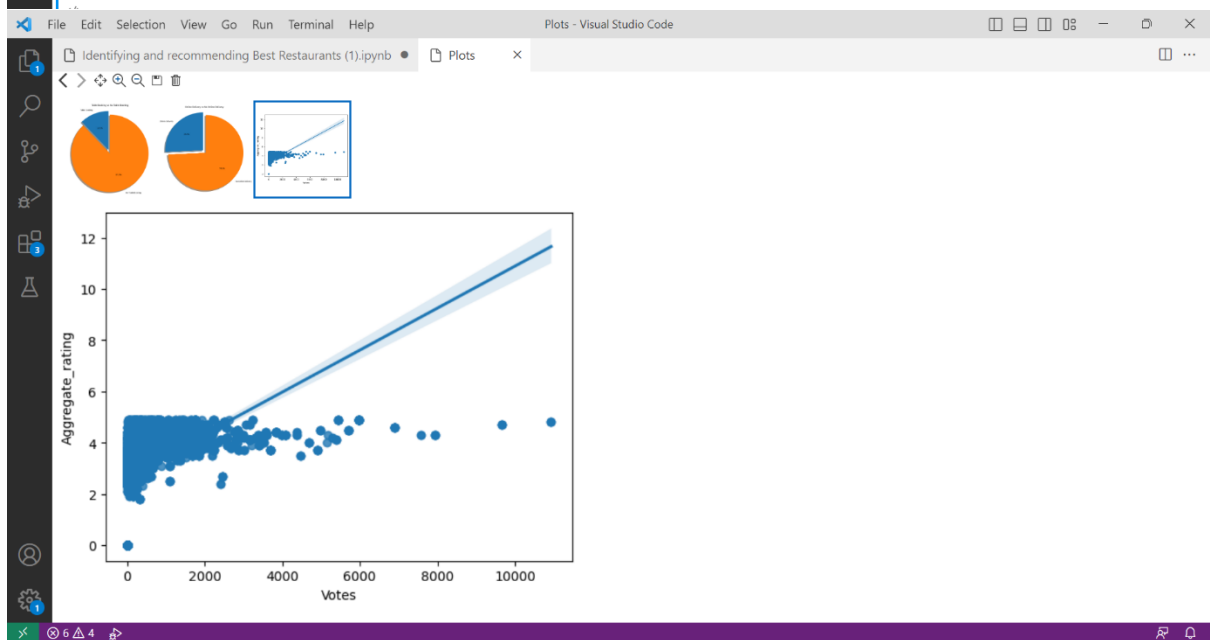
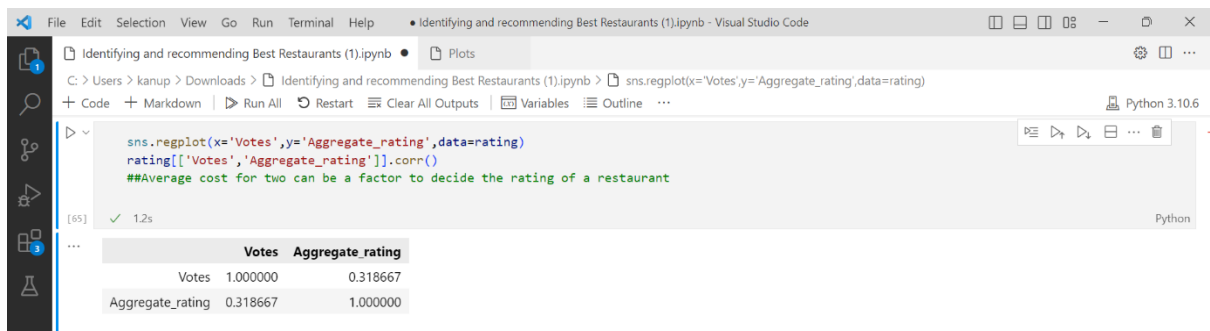
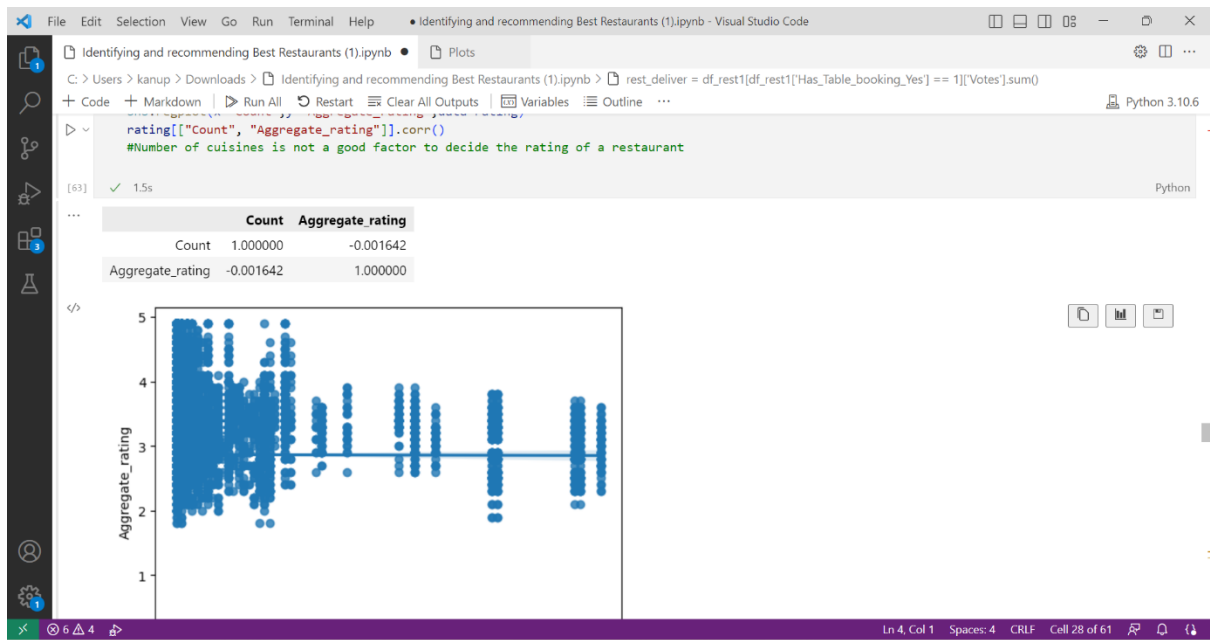
rating = rating.merge(max_rate1,left_on='Restaurant_Name',right_on='Restaurant_Name',how='left')
rating

[62] ✓ 0.0s Python
...

```

	Restaurant_ID	Restaurant_Name	Country	City	Aggregate_rating	Average_Cost_for_two	Votes	Price_range	Has_Table_booking_Yes	Has_Online_delivery
0	7402935	Skye	Indonesia	Jakarta	4.1	800000	1498	3	0	
1	7402935	Skye	Indonesia	Jakarta	4.1	800000	1498	3	0	
2	7410290	Satoo - Hotel Shangri-La	Indonesia	Jakarta	4.6	800000	873	3	0	
3	7410290	Satoo - Hotel Shangri-La	Indonesia	Jakarta	4.6	800000	873	3	0	
4	7410290	Satoo - Hotel Shangri-La	Indonesia	Jakarta	4.6	800000	873	3	0	
...	...	...	...	...	...	...	...	...	...	...
23810	18312106	UrbanCrave	India	Kanpur	3.9	0	127	1	0	

Ln 2, Col 1 Spaces: 4 CRLF Cell 28 of 61



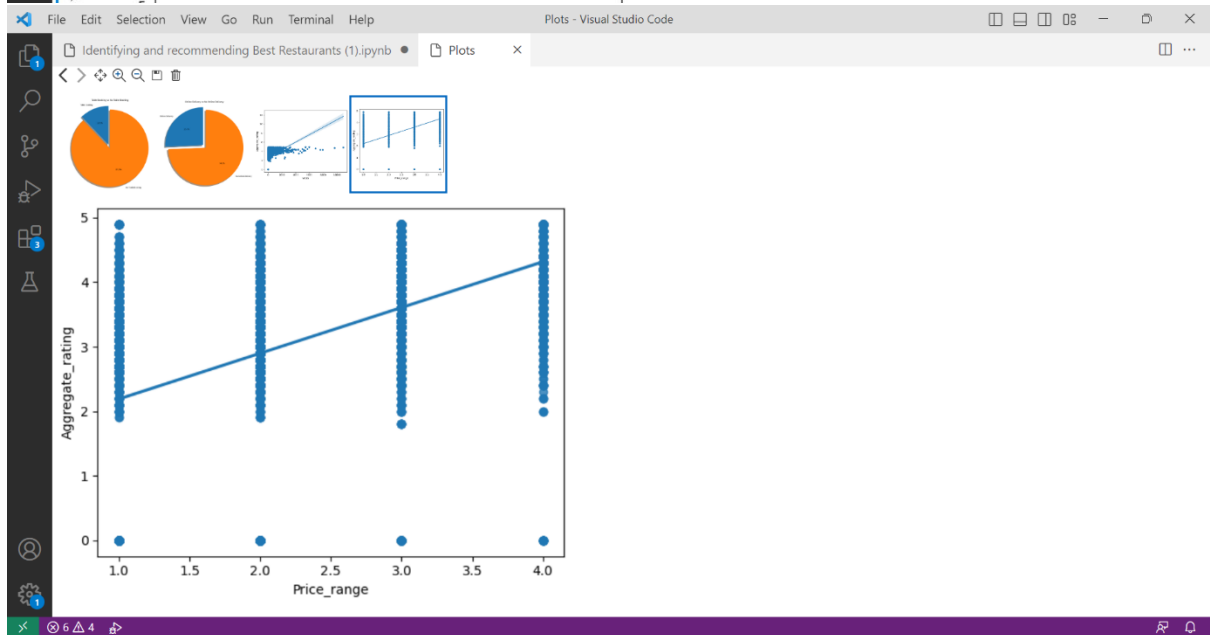
```
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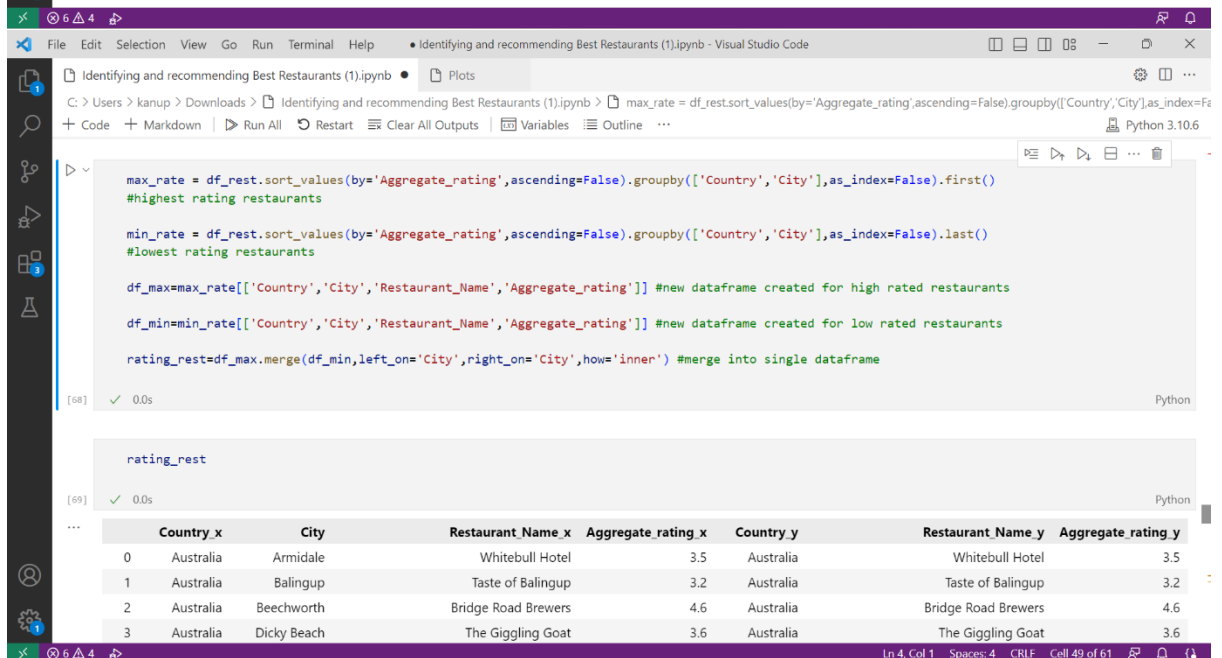
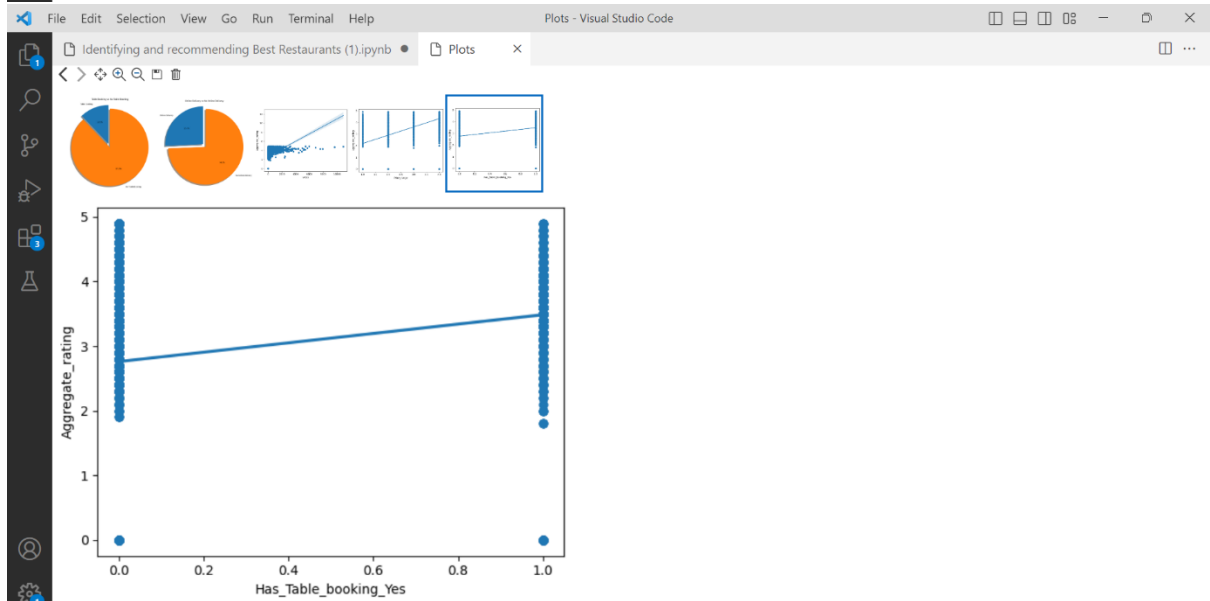
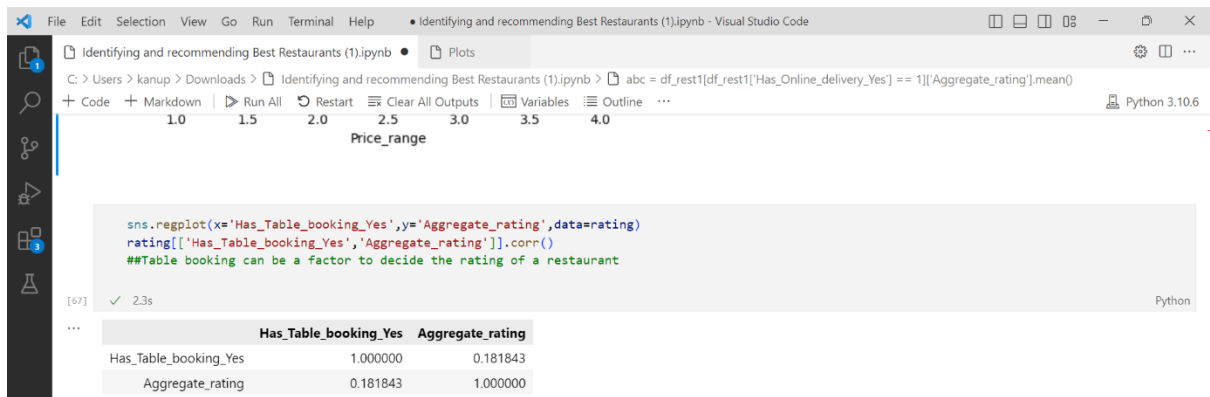
Identifying and recommending Best Restaurants (1).ipynb • Plots
C:\Users\kanup> Downloads > Identifying and recommending Best Restaurants (1).ipynb > abc = df_rest1[df_rest1['Has_Online_delivery_Yes'] == 1]['Aggregate_rating'].mean()
+ Code + Markdown | Run All | Restart | Clear All Outputs | Variables | Outline ... Python 3.10.6

abc = df_rest1[df_rest1['Has_Online_delivery_Yes'] == 1]['Aggregate_rating'].mean()
xyz = df_rest1[df_rest1['Has_Online_delivery_Yes'] == 0]['Aggregate_rating'].mean()
sns.regplot(x='Price_range',y='Aggregate_rating',data=rating)
rating[['Price_range','Aggregate_rating']].corr()
##Price range can be a factor to decide the rating of a restaurant

[56] ✓ 1.7s Python

...
      Price_range  Aggregate_rating
Price_range  1.000000      0.462983
Aggregate_rating  0.462983      1.000000
```





```
File Edit Selection View Go Run Terminal Help • Identifying and recommending Best Restaurants (1).ipynb - Visual Studio Code
C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > max_rate = df_rest.sort_values(by='Aggregate_rating',ascending=False).groupby(['Country','City'],as_index=False)
+ Code + Markdown | Run All | Restart | Clear All Outputs | Variables | Outline ... Python 3.10.6

rating_rest_city_india=rating_rest[rating_rest['Country']=='India'] #storing the dataframe only for country 'India'
rating_rest_city_india #In India
city=rating_rest_city_india['City'].tolist()#converting the series to list
rate_max=rating_rest_city_india['Rating Max'].tolist()#converting the series to list
rate_min=rating_rest_city_india['Rating Min'].tolist()#converting the series to list
rest_name_high=rating_rest_city_india['Highest Rated Restaurant'].tolist()#converting the series to list
rest_name_low=rating_rest_city_india['Lowest Rated Restaurant'].tolist()

[71] ✓ 0.0s Python
```

```
pip install plotly
[72] ✓ 1.7s Python

... Requirement already satisfied: plotly in c:\users\kanup\appdata\local\programs\python\python310\lib\site-packages (5.17.0)
Requirement already satisfied: tenacity>=6.2.0 in c:\users\kanup\appdata\local\programs\python\python310\lib\site-packages (from plotly) (8.2.3)
Requirement already satisfied: packaging in c:\users\kanup\appdata\roaming\python\python310\site-packages (from plotly) (21.3)
Requirement already satisfied: pyparsing!=3.0.5,>=2.0.2 in c:\users\kanup\appdata\roaming\python\python310\site-packages (from packaging->plotly) (3.0)
Note: you may need to restart the kernel to use updated packages.

[notice] A new release of pip available: 22.2.1 -> 23.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

```
import plotly
```

```
File Edit Selection View Go Run Terminal Help • Identifying and recommending Best Restaurants (1).ipynb - Visual Studio Code
C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > max_rate = df_rest.sort_values(by='Aggregate_rating',ascending=False).groupby(['Country','City'],as_index=False)
+ Code + Markdown | Run All | Restart | Clear All Outputs | Variables | Outline ... Python 3.10.6

import plotly.graph_objs as go
plotly.offline.init_notebook_mode()

[81] ✓ 1.0s Python

...

stack0 = go.Bar( # GroupBarChart 1 (Highest Rated Restaurant)
    x=city,#x axis label
    y=rate_max,# y axis label
    text=rest_name_high,# the value of the restaurant
    name='Highest Rated Restaurant',
    marker=dict(
        color='rgb(76,153,0)', #color of the bar graph's marker
        line=dict(
            color='rgb(76,153,0)', #color of the bar graph's line
            width=1.5, #width of the bar graph
        ),
        opacity=1.0
    ),
)
stack1 = go.Bar( # GroupBarChart 2 (Lowest Rated Restaurant)
    x=city,
    y=rate_min,
    text=rest_name_low,
    name='Lowest Rated Restaurant',
    marker=dict(
```

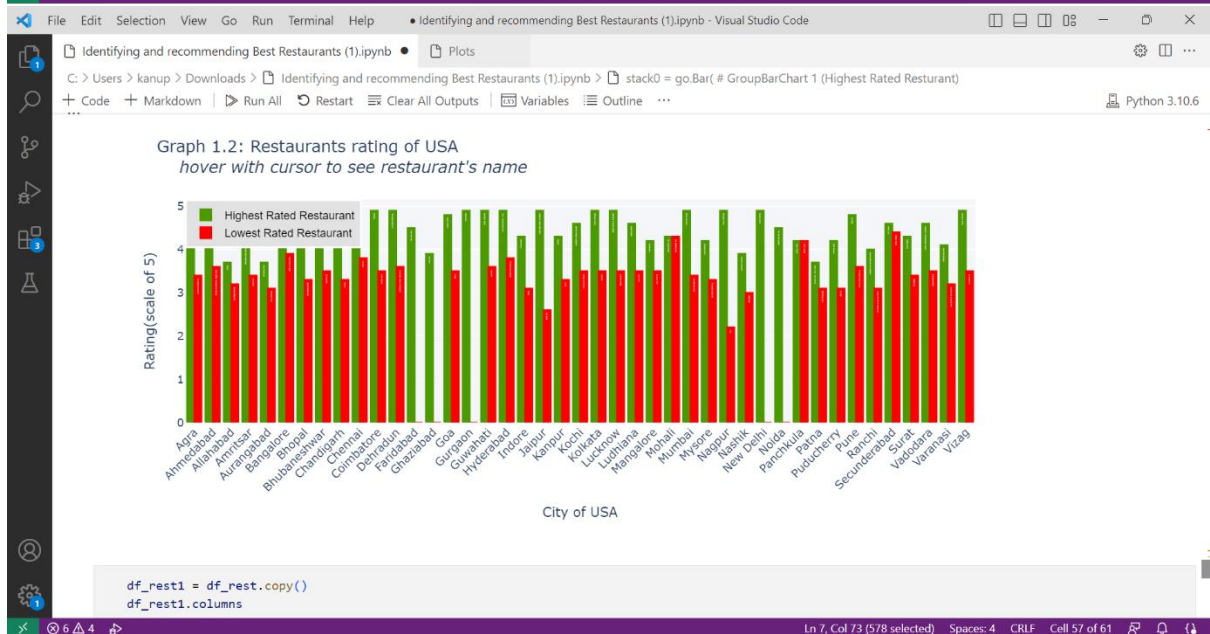
Graph 1.1: Restaurants rating of India
hover with cursor to see restaurant's name



```
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Identifying and recommending Best Restaurants (1).ipynb • Plots
C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > stack0 = go.Bar( # GroupBarChart 1 (Highest Rated Restaurant)
+ Code + Markdown | Run All | Restart | Clear All Outputs | Variables | Outline ... Python 3.10.6

stack0 = go.Bar( # GroupBarChart 1 (Highest Rated Restaurant)
    x=city,#x axis label
    y=rate_max,# y axis label
    text=rest_name_high,# the value of the restaurant
    name='Highest Rated Restaurant',
    marker=dict(
        color='rgb(76,153,0)', #color of the bar graph's marker
        line=dict(
            color='rgb(76,153,0)', #color of the bar graph's line
            width=1.5, #width of the bar graph
        )
    ),
    opacity=1.0
)

stack1 = go.Bar( # GroupBarChart 2 (Lowest Rated Restaurant)
    x=city,
    y=rate_min,
    text=rest_name_low,
    name='Lowest Rated Restaurant',
    marker=dict(
        color='rgb(255,0,0)', #color of the bar graph's marker
        line=dict(
            color='rgb(255,0,0)', #color of the bar graph's line
            width=1.5, #width of the bar graph
        )
    ),
    opacity=1.0
)
```



```
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C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > stack0 = go.Bar( # GroupBarChart 1 (Highest Rated Restaurant)
+ Code + Markdown | Run All | Restart | Clear All Outputs | Variables | Outline ... Python 3.10.6

df_rest1 = df_rest.copy()
df_rest1.columns

[87] ✓ 0.0s Python
...
Index(['Restaurant_ID', 'Restaurant_Name', 'Country_Code', 'City', 'Address',
      'Locality', 'Locality_Verbose', 'Longitude', 'Latitude', 'Cuisines',
      'Average_Cost_for_two', 'Currency', 'Has_Table_booking',
      'Has_Online_delivery', 'Price_range', 'Aggregate_rating',
      'Rating_color', 'Rating_text', 'Votes', 'Country'],
      dtype='object')

dummy = ['Has_Table_booking', 'Has_Online_delivery']
df_rest1 = pd.get_dummies(df_rest1, columns=dummy, drop_first=True)
df_rest1.head()
# 0 indicates 'NO'
# 1 indicates 'YES'

[88] ✓ 0.0s Python
...
Restaurant_ID Restaurant_Name Country_Code City Address Locality Locality_Verbose Longitude Latitude Cuisines Average_Cost_for_two
0 7402935 Skye 94 Jakarta Menara BCA, Lantai 56, Jl. MH. Thamrin, Grand Indonesia Mall, Grand Indonesia Mall, Thamrin, Jakarta 106.821999 -6.196778 Italian, Continental 800000
R
```

```
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Identifying and recommending Best Restaurants (1).ipynb • Plots
C: > Users > kanup > Downloads > Identifying and recommending Best Restaurants (1).ipynb > stack0 = go.Bar( # GroupBarChart 1 (Highest Rated Restaurant)
+ Code + Markdown | Run All | Restart | Clear All Outputs | Variables | Outline ... Python 3.10.6

dummy = ['Has_Table_booking', 'Has_Online_delivery']
df_rest1 = pd.get_dummies(df_rest1, columns=dummy, drop_first=True)
df_rest1.head()
# 0 indicates 'NO'
# 1 indicates 'YES'

[88] ✓ 0.0s Python
...
sines Average_Cost_for_two Currency Price_range Aggregate_rating Rating_color Rating_text Votes Country Has_Table_booking_Yes Has_Online_delivery_Yes
alian, ental 800000 Indonesian Rupiah(IDR) 3 4.1 Green Very Good 1498 Indonesia 0 0
sian, sian, stern 800000 Indonesian Rupiah(IDR) 3 4.6 Dark Green Excellent 873 Indonesia 0 0
iushi, inese 500000 Indonesian Rupiah(IDR) 3 4.9 Dark Green Excellent 605 Indonesia 0 0
inese 450000 Indonesian Rupiah(IDR) 3 4.2 Green Very Good 395 Indonesia 0 0
```